

PathMoE-MixNet Methodology Flow Diagram

Stage	Description
1. Dataset	BREAKHIS (40x/100x/200x/400x)
2. Quality Control	Corrupted image removal
3. Preprocessing	Macenko stain normalization
4. Augmentation	Rotation, Flip, Crop, Color Jitter, MixUp, CutMix, Random Erasing
5. Patch Extraction	224×224 patches
6. Parallel Branches	EfficientNetV2+CBAM MaxViT/RI-ViT MLP-Mixer
7. Fusion	Cross-attention + Learnable weighting + Residual fusion
8. Sparse MoE	Router → CNN/ViT/Mixer experts (Top-k)
9. Cross-Magnification	Scale-aware attention + aggregation
10. Explainability	Grad-CAM++, Attention Rollout, Expert visualization
11. Classification	Linear → BN → Dropout → Linear
12. Uncertainty	Conformal prediction
13. Output	8-class prediction + confidence + heatmap + uncertainty

Flow:
Dataset → Quality Check → Stain Normalization → Augmentation → Patch Extraction → Parallel CNN/Transformer/MLP-Mixer → Adaptive Fusion → Sparse MoE → Cross-Magnification Fusion → Explainability → Classification → Conformal Prediction → Final Diagnosis